PRESENTING

FRUIT-FLY SPY

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Project Prompt #1:

- Gather various datasets that provide insight into how fruit flies spread throughout the United States
- Create a method of aggregating data into a meaningful risk factor
- Provide a method of visualizing data in a way that is easy to understand

TOOLS USED

PANDAS

FOLIUM+GEOPY

ARCGIS

HOW WE SOLVED IT:

For the solution,

- Parsed csv files from pandas & geopandas function
- Utilized GIS import to find county that fruit flies were located

STEPS:

- We started off with planning how we would execute this project
- We listed the portions of the program that we would have to create into a checklist
- **Gathering Data**
- Aggregate meaningful parts of the collected data into a smaller dataset
- Use the dataset to determine a risk factor

ArcGIS

- Tried interacting with ArcGIS Online - Very Slow
- Tried exporting data from ArcGIS Pro - Failed to export flight data
- Used arcpy to interact with the downloaded ArcGIS layers
    - Faster, but still the slowest part
    - Lots of time troubleshooting and trying to understand documentation
- Extracted detection numbers and international flights

Census Bureau

- Created a script to utilize provided census APIs to extract data regarding imports
    - Takes a while
    - Narrowed data using HS6 codes which apply to fruit flies
    -

IPPC

- Scraped month-by-month data with keywords to identify postive and negative reports of pest presence

EPPO Global Database

- Gathered data on the 3 target species - Oriental, Mexican, Mediterranean Fruit Flies
- Indicated where fruit flies were present outside the US

Order data by the following

```
{
   City: {
      MM/YYYY: {
         PassengerData: ...
         CargoData: ...
         ConfirmedPresence: ... (in US)
            RiskCountries: ... (All countries which were an origin of cargo or passengers and contain a fruit fly risk, ordered by Country → Species)
      }
   }
}
```

Parse new data, determine a risk factor based on the number of risk countries and volume of cargo/passengers

Intended to add:

- Weather
- Crop Data
- Domestic flights

Export as City → MM/YYYY → Species → RiskLevel

User Interface

- First tried to use ArcGIS-Mapping without success
- Chose to use Folium due to it's adjacency to ArcGIS and ability to handle shapefiles.
- Using ArcGIS' API, generated hot zones on a publicly available shapefile of the US.
- Used risk factors to shade certain zones redder than others.

THANK YOU!